CHAPTER 3

Complex numbers and trigonometry

Addition formulas, exponential form, de Moivre's and Euler's formulas

Objectives. Establish the addition and double-angle formulas; write a complex number in exponential form and choose the form suited to a computation; compute powers; use de Moivre's and Euler's formulas to transform trigonometric expressions.

Prerequisites. Chapter 2; scalar product (Première); algebraic properties of the exponential function; unit circle and associated angles.

Fluency

Fluency · No calculator, five minutes.

1. Give $\cos\frac{\pi}{6}$ and $\sin\frac{5\pi}{6}$.
2. Express $\cos(\pi - x)$ and $\sin\left(\frac{\pi}{2} - x\right)$ in terms of $\cos x$ or $\sin x$.
3. Recall two expressions of the scalar product $\vec{a} \cdot \vec{b}$.
4. Simplify $\mathrm{e}^{a} \times \mathrm{e}^{b}$ and $(\mathrm{e}^{a})^{n}$.
5. Solve in $(-\pi, \pi]$ the equation $\cos x = \frac{1}{2}$.
6. Give the modulus and an argument of $-1 - i$.